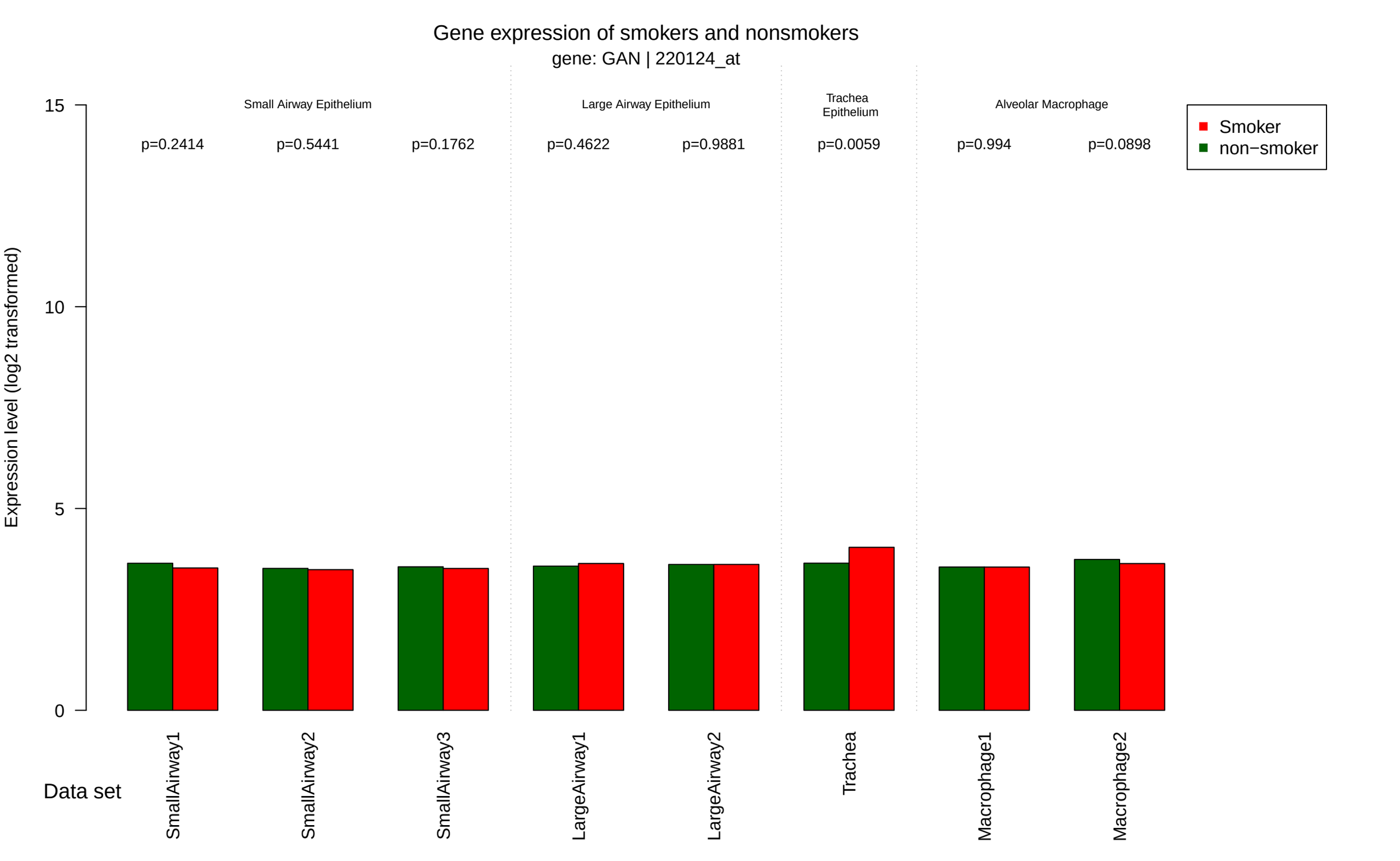
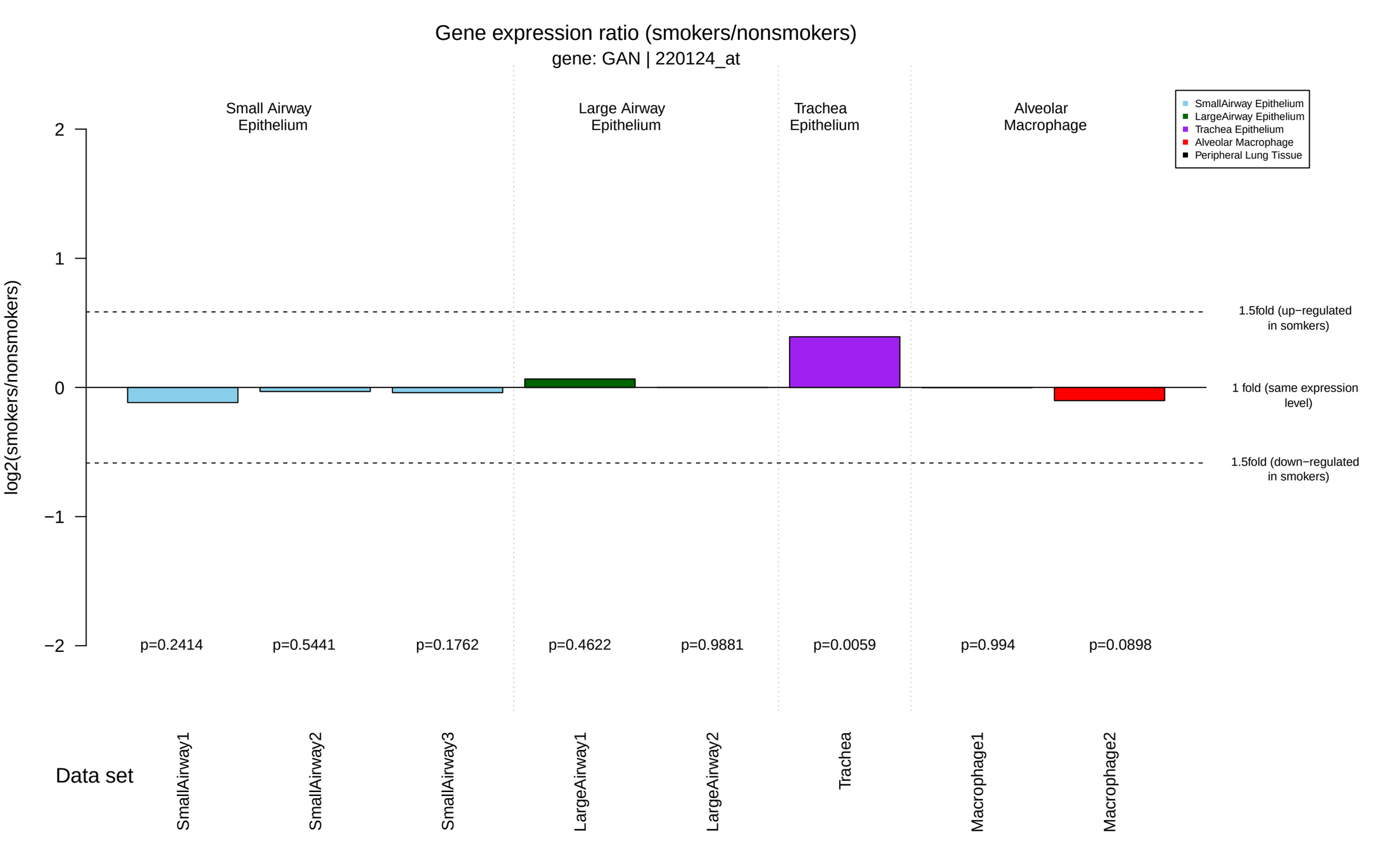
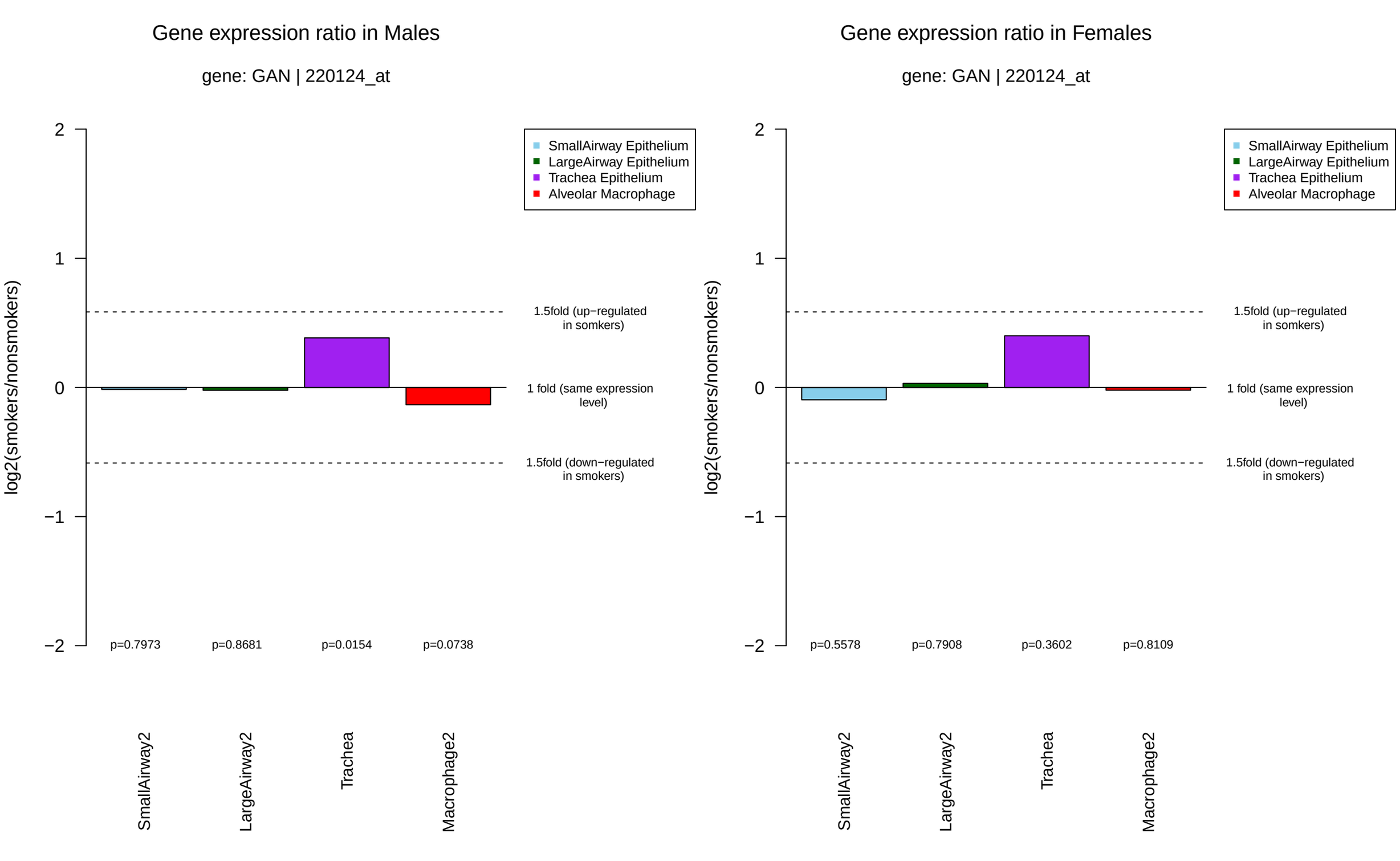
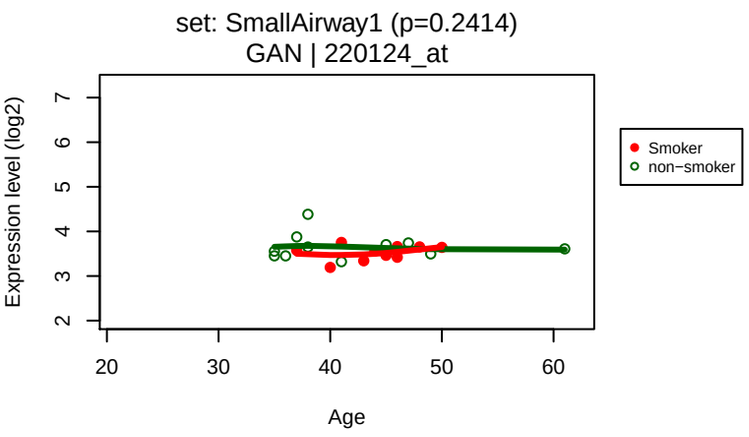
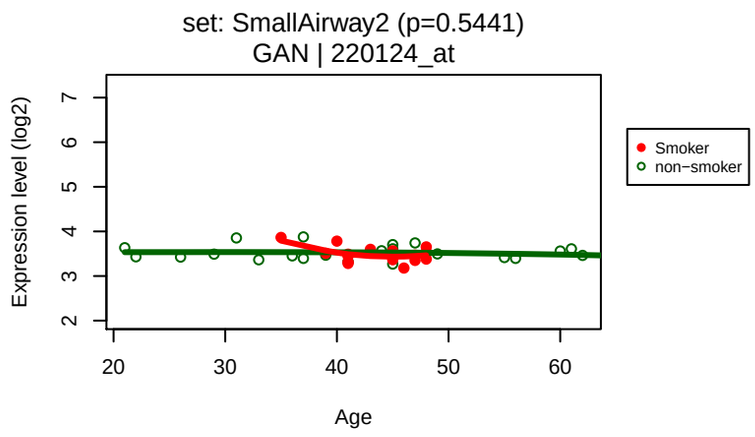
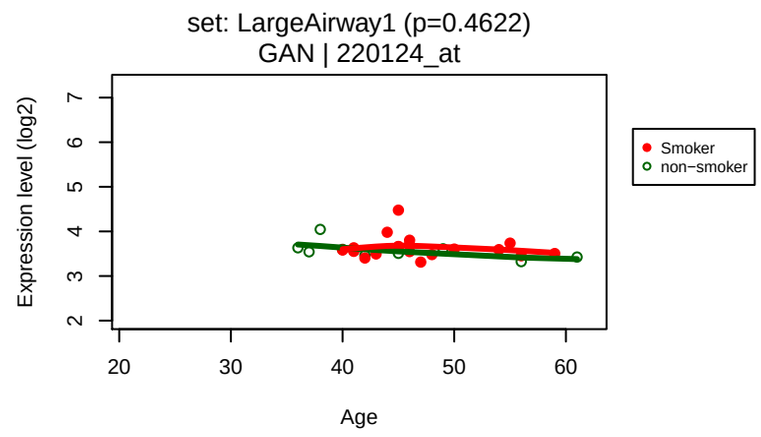

Gene expression of smokers and nonsmokers
gene: GAN | 220124_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2414
p=0.5441
p=0.1762
p=0.4622
p=0.9881
p=0.0059
p=0.994
p=0.0898
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

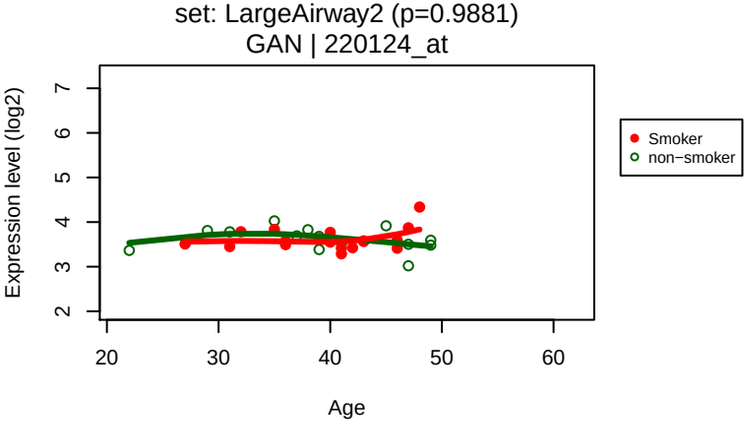

Gene expression ratio (smokers/nonsmokers)
gene: GAN | 220124_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2414
p=0.5441
p=0.1762
p=0.4622
p=0.9881
p=0.0059
p=0.994
p=0.0898
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

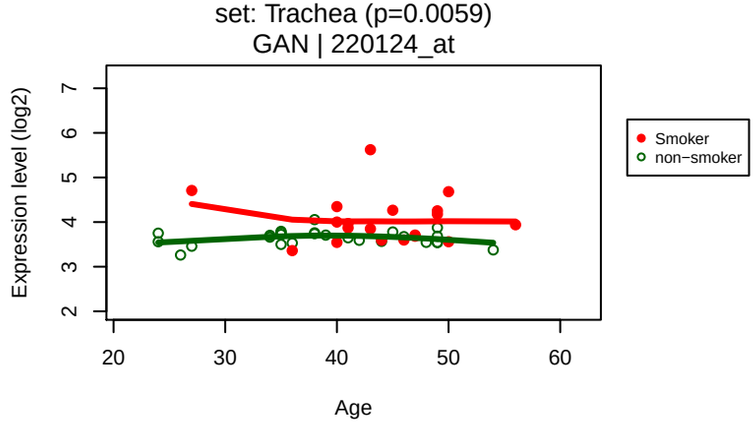

Gene expression ratio in Males
gene: GAN | 220124_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7973
p=0.8681
p=0.0154
p=0.0738
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAN | 220124_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5578
p=0.7908
p=0.3602
p=0.8109
SmallAirway2
LargeAirway2
Trachea
Macrophage2

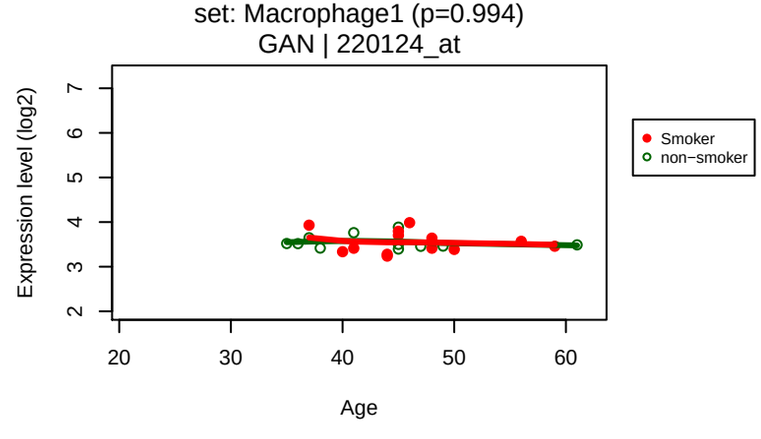

set: SmallAirway1 (p=0.2414)
GAN | 220124_at
Expression level (log2)
Age
Smoker
non−smoker

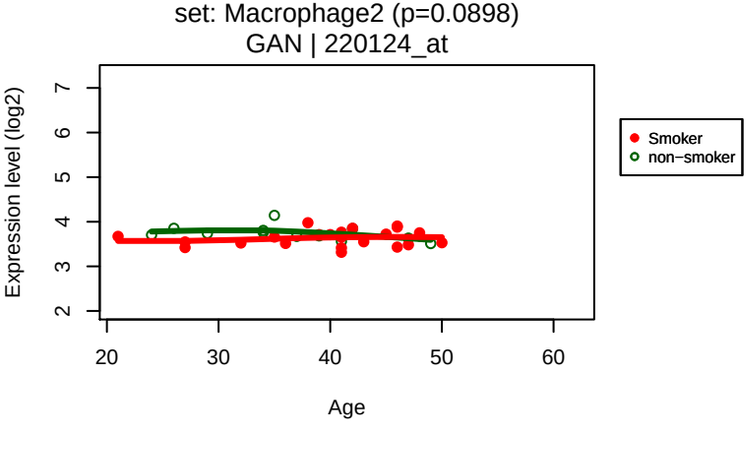

set: SmallAirway2 (p=0.5441)
GAN | 220124_at
Expression level (log2)
Age
Smoker
non−smoker

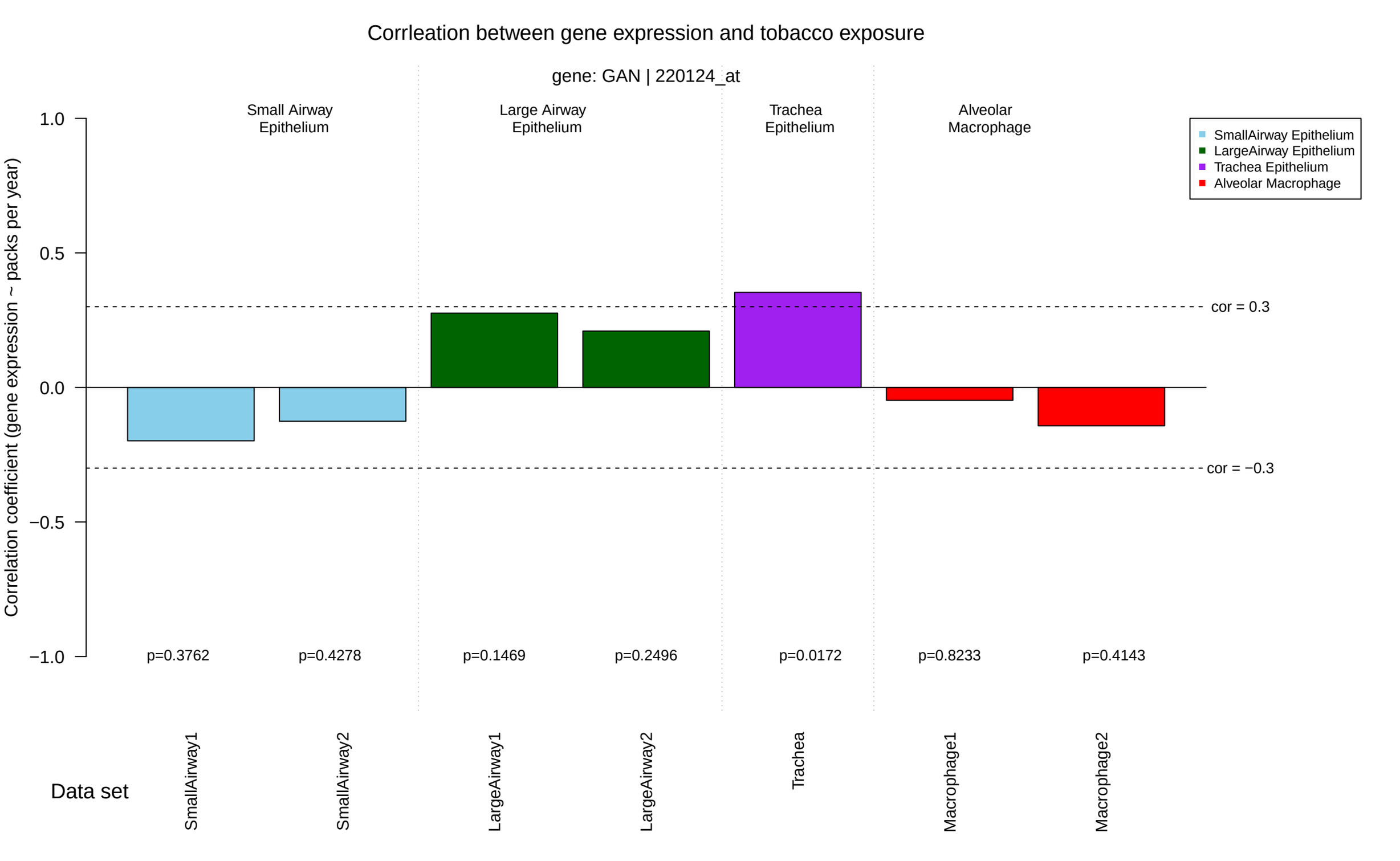

set: LargeAirway1 (p=0.4622)
GAN | 220124_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9881)
GAN | 220124_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0059)
GAN | 220124_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.994)
GAN | 220124_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0898)
GAN | 220124_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAN | 220124_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3762
p=0.4278
p=0.1469
p=0.2496
p=0.0172
p=0.8233
p=0.4143
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2